NETWORK ATTACK DETECTION
AND
INVESTIGATION USING WIRESHARK
in SOC LAB ENVIRONMENT

Introduction:

This project focuses on simulating and analyzing common network reconnaissance activities in a controlled home lab environment. The purpose of this lab was to understand how early-stage attack behavior appears at the packet level and how such activity can be detected from a Security Operations Center (SOC) perspective using Wireshark.

The project emphasizes practical exposure to traffic monitoring, protocol analysis, & identifying suspicious patterns in network communication.

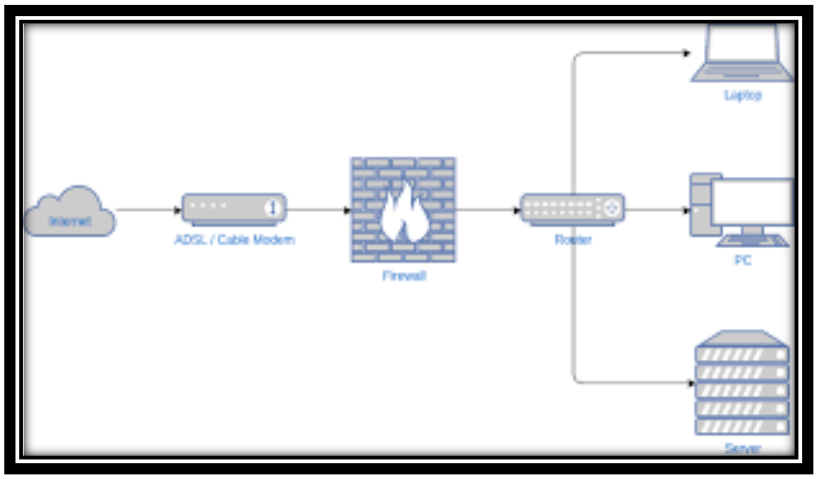

fig: Basic Network Diagram

Objective:

The primary objective of this project is:

a) To simulate basic network reconnaissance techniques.
b) To analyze the ICMP-based host discovery traffic.
c) To perform & monitor TCP SYN port scanning using Nmap.
d) To understand how such activities appear in packet capture.
e) To develop analytical thinking from a SOC perspective.

Lab Setup / Tools used: Lab environment config:

1. Attacker Machine:- Kali Linux (Virtual Machine) was configured as the attacker system to simulate reconnaissance and port scanning activities against the target machine.
2. Target Machine:- Windows 11 was configured as the target system to receive and respond to simulated attack traffic generated from the Kali Linux machine.
3. Network Configuration: Under this, either Bridged Adapter / Host-only-network is used on Kali Linux (VM) for communication with each other properly.
4. Network Monitoring tool:- Wireshark was used to capture, inspect, & analyze real-time network traffic at the packet level. It was not used for scanning, but for observing & investigating network comms generated during, simulated attacks.
5. Scanning tool: Nmap (Network Mapper) was used as the active scanning tool to perform host discovery and TCP SYN port scanning against the target system.

Both virtual machines were configured within the same network range to enable successful communication before executing attack simulations.

Simulation:

2 Different attack scenarios were simulated in the lab environment to understand how common cyber-threats appear in the network.

1. **Ping Sweep:** In this scenario, a basic reconnaissance technique was simulated where the attacker verifies whether the target system is active on the network using ICMP Echo Requests.

   Purpose: The main purpose of this attack test is to understand how host discovery appears from a SOC monitoring perspective.

COMMAND USED: ping <Windows IP>

Attack Screenshot:

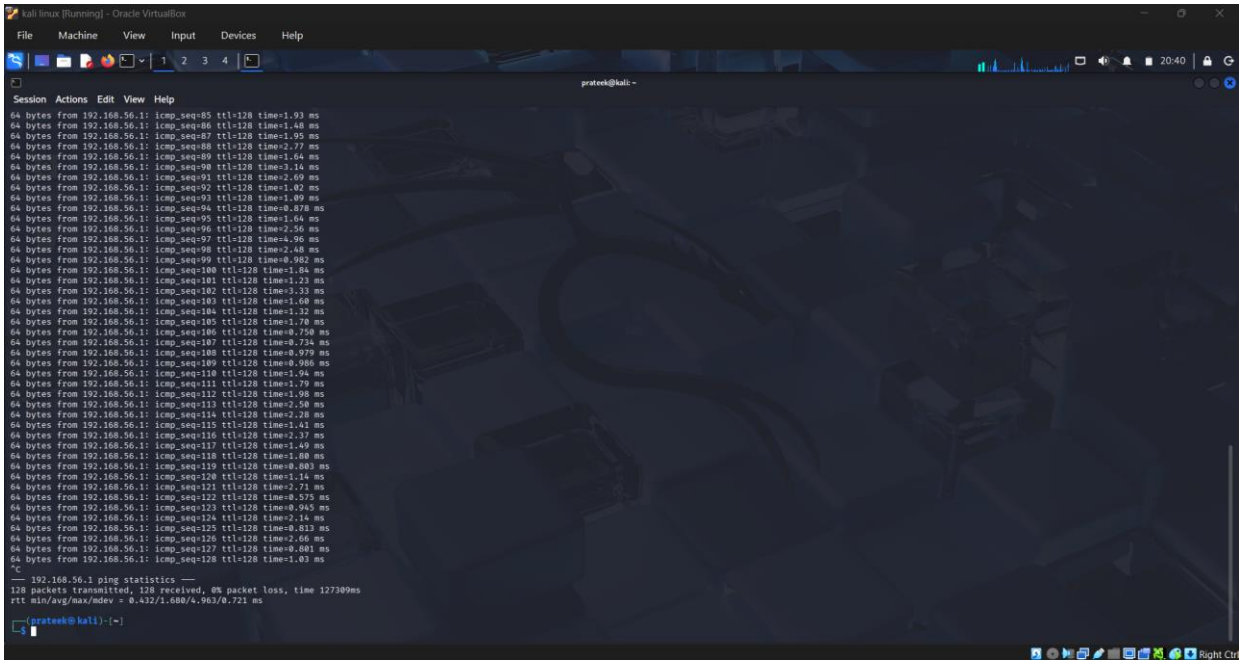

fig 1.1: Attempting the Attack of Ping Sweep from Kali VM.

Wireshark Capturing:

- **Wireshark:-** As said, this is a type of Network Scanning tool which is used to capture, troubleshoot, & inspect the network traffic in real-time.
- **ICMP:** The display filter icmp was applied in Wireshark to isolate and analyze only ICMP packets, specifically Echo Requests & Replies generated during the ping sweep test.

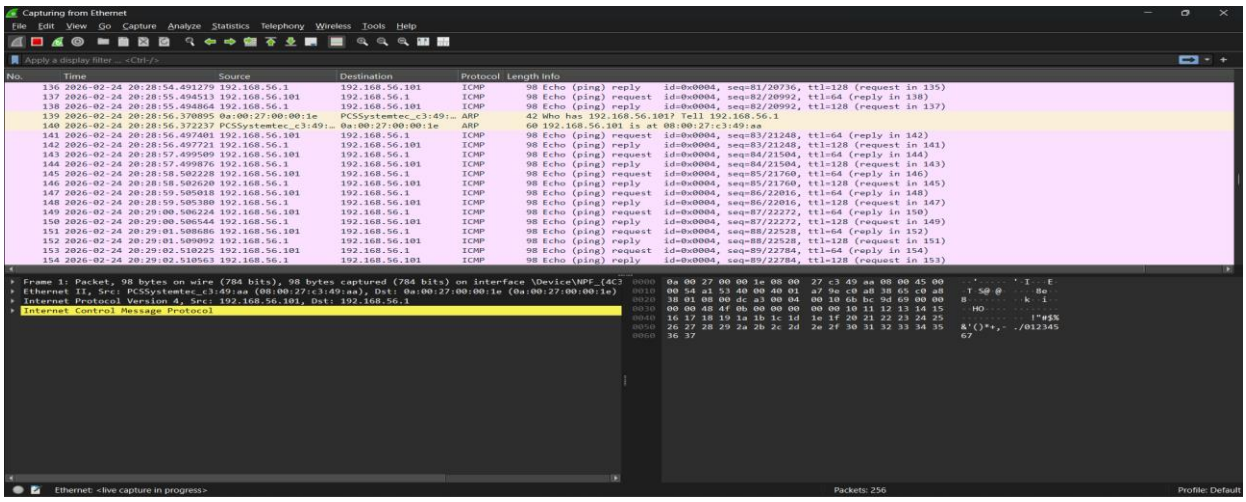

fig 1.2: ICMP Echo Request & Reply Captured in Wireshark.

Observed Behavior:

- Continuous ICMP Echo requests were observed originating from the Kali linux.
- Corresponding ICMP Echo replies from Windows system.
- The communication occurred at regular one second intervals.
- No packet loss observed during testing.

This confirmed successful bidirectional ICMP Communication.

SOC Perspective: From the SOC perspective:-

Repeated ICMP requests may indicate host discovery (or) network reconnaissance activity. While occasional ping traffic is normal behavior, systematic or large-scale ICMP probing across multiple IP addresses can signal early-stage attacker reconnaissance.

2. Port Scanning using Nmap:

Port scanning is basically a type of technique which attackers use to:

- ✓ Discover open ports.
- ✓ Identify running services.
- ✓ Find potential entry points.

A TCP SYN stealth scan (-sS) was performed to simulate real-world reconnaissance behavior typically used by attackers during the enumeration phase.

Nmap: Nmap is an active network scanning tool that sends specially crafted packets to target hosts to discover open ports, running services, potential exposure too.

Command Used:

a) nmap <Windows IP>.
b) nmap -sS <Windows IP> -- SYN Stealth Scan.
c) In Wireshark filter -- tcp.flags.syn == 1 && tcp.flags.ack == 0.

Attack Command Screenshot:

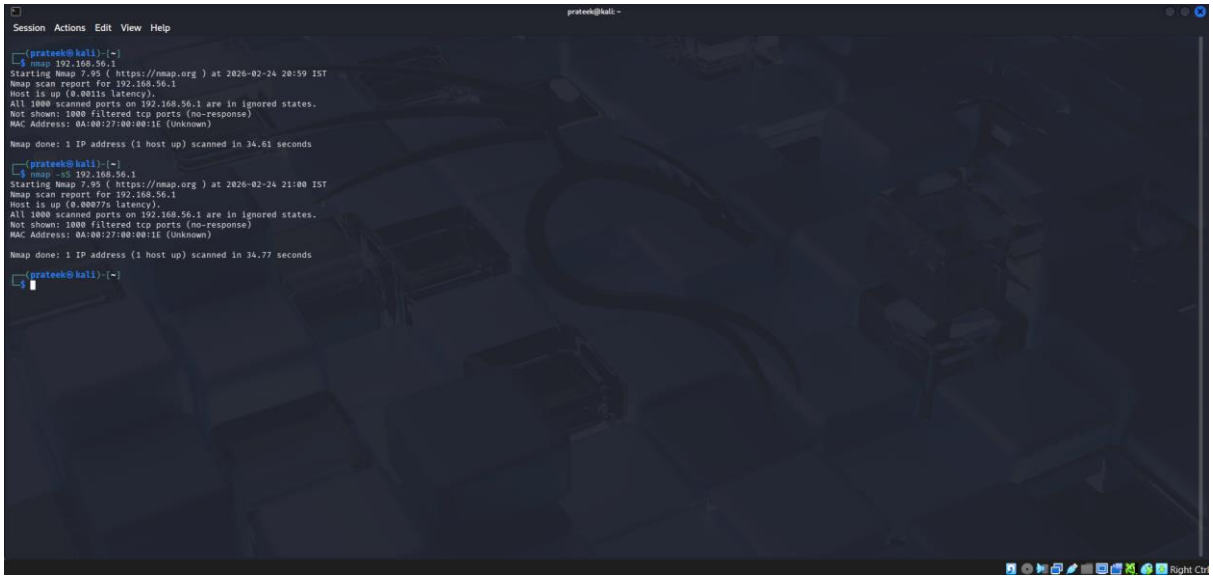

fig 2.1: Using the Nmap tool to start the attack.

Wireshark Capturing:

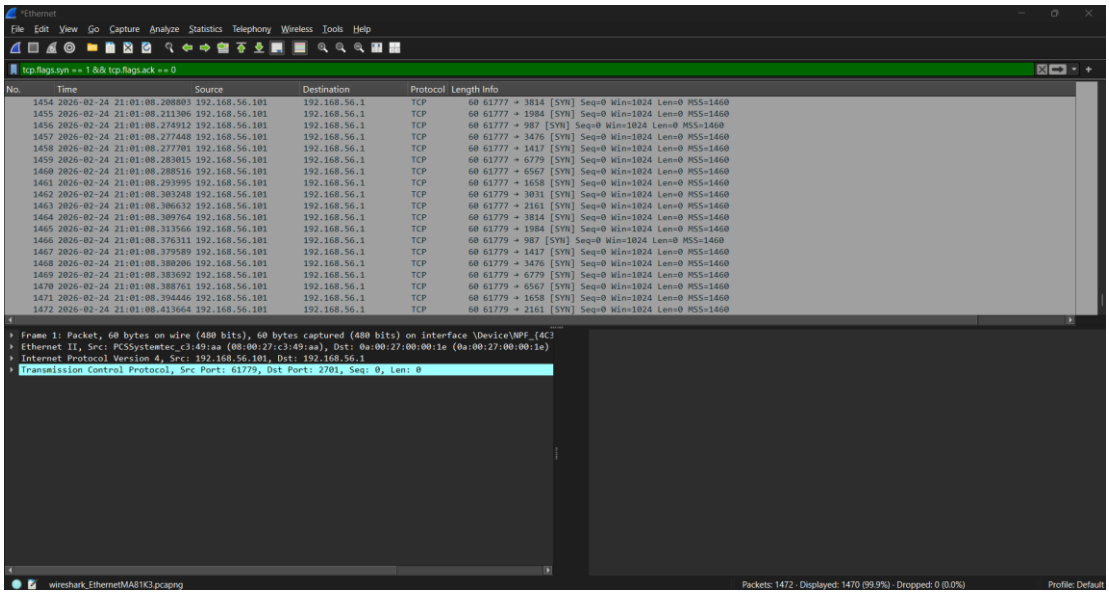

fig 2.2: TCP SYN Packets observed during Nmap Scan .

Observations:

- Multiple TCP SYN packets were observed originating from the Kali Linux IP Address.
- The destination ports varied continuously while the source IP remained constant.
- Closed ports responded with TCP RST (Reset) packets.
- Open ports (if any), responded with SYN-ACK packets.
- Most connections did not complete the full TCP three-way handshake, indicating scanning behavior rather than legitimate session establishment.
- A noticeable spike in packet volume occurred during the scan duration.
- The scan behavior matched TCP SYN scanning methodology used by Nmap.

SOC Perspective: The observed traffic pattern clearly indicates active reconnaissance through TCP SYN scanning. This behavior is typically associated with pre-exploitation enumeration, where attackers attempt to identify exposed services before launching targeted attacks.

In real-world environments, such activity is commonly flagged by IDS/IPS or SIEM solutions as suspicious port scanning behavior.

Indicators of Compromise (IOCs) Identified:

- A High volume of SYN packets within a short duration.
- Multiple destination ports targeted sequentially.
- Incomplete TCP handshakes.
- A single source IP address probing multiple destination ports and services.

Risk Assessment:

The detected activity is classified as **Medium Risk**, as it indicates active reconnaissance. While no exploitation was observed, such scanning behavior often precedes targeted attacks.

Conclusion:

This project provided hands-on experience in identifying and analyzing early-stage reconnaissance techniques using Wireshark. By simulating ICMP-based host discovery and TCP SYN port scanning, I gained a deeper understanding of how attackers perform enumeration and how such behavior can be detected from a SOC analyst's perspective.

The lab strengthened my ability to interpret packet-level data and differentiate between normal network traffic and suspicious activity.

7.